

(a) The genus *Leopardus* consists of species of wild cats that are small and spotted. In 2013, biologists investigated the evolution of *Leopardus tigrinus* in South America.

A map of South America with two regions highlighted. Region A is an oval in the northeast, labeled 'population A'. Region B is an irregular shape in the south, labeled 'population B'.

- Population **A** lives in grassland and desert habitats and population **B** lives in forest habitats.
- The cats in population **A** have a lighter coat colour and a different pattern of spots from the cats in population **B**.
- Genetic analysis shows that population **A** is genetically distinct from population **B**.
- Population **B** has now been reclassified as a new species, *L. guttulus*.

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- (b) Illegal trade threatens the survival of endangered species. Forensic tests can identify animal species from the DNA of their body parts, for example bones.

Cat species differ in the number of repeats of bases C and A (CA repeats) within one particular region of DNA, known as Ple46.

Table 3.1 shows the number of repeats in Ple46 for three endangered cat species.

Table 3.1

cat species	number of CA repeats in Ple46
Asiatic lion	22
Bengal tiger	7–8
leopard	14–15

- (i) Gel electrophoresis is used to estimate the length of Ple46 in a sample of DNA to check whether the sample comes from one of the endangered cat species in Table 3.1.

Outline how gel electrophoresis is carried out to confirm whether the sample comes from one of these endangered cat species.

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- (ii) Customs officials are responsible for checking whether animal body parts come from species on the CITES list of endangered species. Officials can now use a small DNA barcoding kit linked to a computer database to identify a species.

Suggest **two** advantages of using this method to identify a species.

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[Total: 12]